

# Real-Time Linux Applicability for Hard Real-Time Systems

Let's get an in-depth understanding of real-time systems, analysing the real-time characteristics of two prominent operating systems: Linux with PREEMPT\_RT patch and QNX. By examining their real-time characteristics, determinism, performance, and features, developers can get an idea of how the real-time behaviours of Linux with PREEMPT\_RT patch compare with QNX.



**R**eal-time operating systems (RTOS) play a critical role in the development of hard real-time systems and meeting the stringent timing requirements of real-time applications. QNX (Quantum Neutrino eXecutive) stands out as a preferred choice for hard real-time systems, such as medical systems, due to its real-time capabilities and features.

Linux, in its standard form, is not considered a hard real-time operating system. However, efforts have been made to make Linux more suitable for real-time applications. Linux with PREEMPT\_RT patch attempts to enhance the real-time capabilities of the standard Linux kernel by adding specific real-time scheduling policies, reducing interrupt latency, and providing mechanisms for prioritising time-critical tasks. These improvements have made Linux with PREEMPT\_RT patch more predictable and responsive.

## Business challenges

A real-time operating system is specialised, designed to meet the stringent timing requirements of hard

real-time systems. These systems often have critical applications in industries such as aerospace, automotive, industrial automation, and medical devices. Nowadays, medical industries are migrating from (proprietary) RTOS such as QNX to Linux (open source) drawn by licence, community support, security, and cost considerations. Linux with strong real-time capabilities is required for this scenario, and for understanding how PREEMPT\_RT patch improves real-time capabilities.

Figure 1 lists some business challenges associated with using RTOS.

## Problem statement

RTOS like QNX are used and deployed in real-time systems in

critical industries, such as medical devices. Industries considering migrating from RTOS to Linux need a detailed understanding of Linux with the PREEMPT\_RT patch -- its real-time capabilities, behaviour, and features. Additionally, a comparison between Linux with PREEMPT\_RT patch and QNX is necessary. The problem is that this information and comparison results are not available. This article will address this issue comprehensively. We also do not know how to use PREEMPT\_RT patch and the real-time behaviours, features and latency reduction offered by it. This issue will be addressed here with PREEMPT\_RT patch guidelines and case studies.

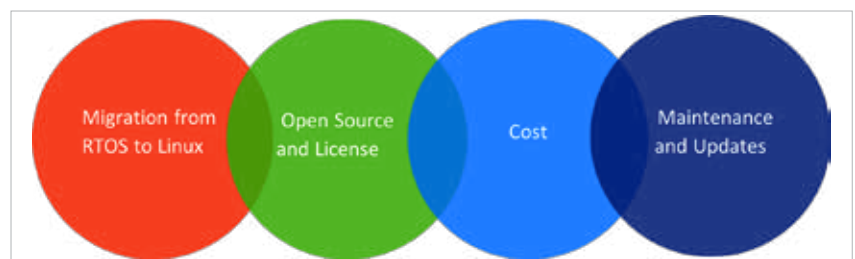


Figure 1: Business challenges